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BAKU HIGHER OIL SCHOOL

CE-403: Database Administration (PostgreSQL)

Elchin Salahzada

Lecture 1

CE-403: Database Management Systems
Baku Higher Oil School : Computer Engineering

Agenda

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- Course Syllabus
- Course Overview
- Course Material
- Grading Policy
- Introduction to DBMS

Course Syllabus

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- Introduction to Databases & DBMS
- SQL Fundamentals I – DDL, DML, basic SELECT, WHERE, ORDER BY, operators
- SQL Fundamentals II – Joins (inner, outer, self), subqueries, set operations
- Aggregate Functions & Grouping
- Database Design & ER Modeling
- Indexes & Query Optimization
- Stored Procedures, Functions & Triggers – Syntax, use cases, best practices
- Transactions & Concurrency Control – ACID properties, isolation levels, locking, deadlocks
- Backup & Recovery – Backup types, restore strategies, point-in-time recovery
- Replication & High Availability – Primary-replica setup, failover, clustering basics
- Database Security – Authentication, RBAC, encryption, SQL injection prevention
- Final Review – Course recap, Q&A, comprehensive practice exam

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Course Overview

- The course teaches you how to be junior level of DBA

- This is not a course on:
- – Senior-level or enterprise database administration (e.g., large-scale architecture, high availability, advanced performance tuning).

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Course Material

- **"Learning PostgreSQL" by Salahaldin Juba, Achim Vannahme, Andrey Volkov** — good starting point covering fundamentals and basic administration.
- **"PostgreSQL: Up and Running" by Regina Obe & Leo Hsu** — practical, concise introduction covering core features and common use cases.
- **"PostgreSQL Administration Cookbook" by Simon Riggs & Gianni Ciolli** - task-oriented solutions for installing, configuring, maintaining, and troubleshooting PostgreSQL databases.

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Grading

Class Activity	10%
Midterm	30%
Lab Assignment	20%
Final Project	40%

LAB Assignment

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- LAB Assignment is used to be sure that you've understood the topic and will have no problem in exams.
- Deadlines for assignments will be set to 5-7 days from announcement.
- It is highly recommended to start working on assignments on the announcement day.
- For each missed deadline day you will get 25% penalty per day.
- Assignments are individual work, so solutions must be individual.
- You may use search engines however, using AI tools are not allowed.

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Class Activity

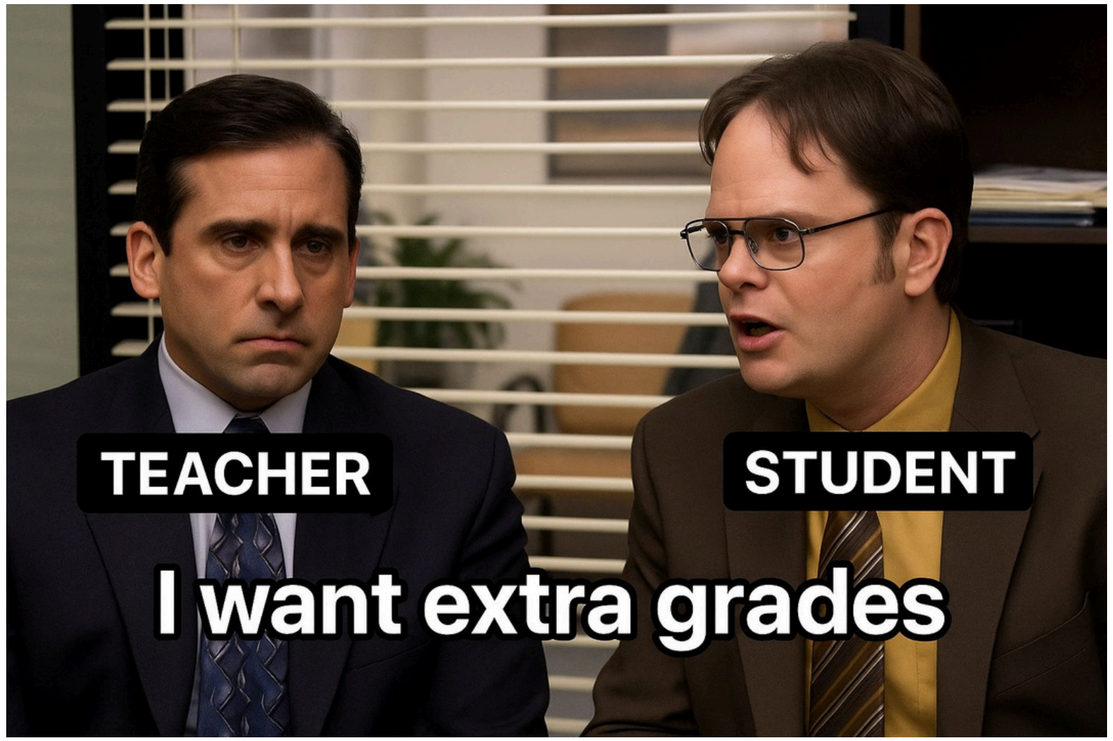
- During lab sessions, students must practice the corresponding lecture material.
- Students' class activity will be graded daily during lectures.
- Students may get better grades, if they:
  - Asking relevant questions regarding lecture
  - Answering questions
  - Any other lecture and lab related class activity

- Misbehavior in class will reduce your daily activity points.
- Examples of misbehavior include:
  - Using computers for non-class purposes
  - Using personal devices during lectures (phone, personal laptop)
  - Talking with classmates during lectures and labs
  - Any behavior irrelevant to class activities will not be tolerated

Don’t

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- No asking for extra points, adjustments, curve, etc.
- If you need higher GPA, think about it now.



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Plagiarism

- Plagiarism in any form is strictly prohibited. The following actions constitute plagiarism:
  - Copying solutions from other students (all parties involved will be held equally responsible)
  - Using AI-generated content
  - Copying material from internet sources without proper attribution

- Plagiarism is detected through specialized software (TurnitinAI Detector).
- Any confirmed violation will result in a 5-point deduction from your Class Activity grade and corresponding assignment will be graded as 0.

Database Management Systems (DBMS)

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- Introduction to DBMS
- Types of Databases
- DBMS Architecture
- The Relational Model
- Keys
- ACID Properties
- Popular DBMS
- Roles in Database World
- Introduction to SQL

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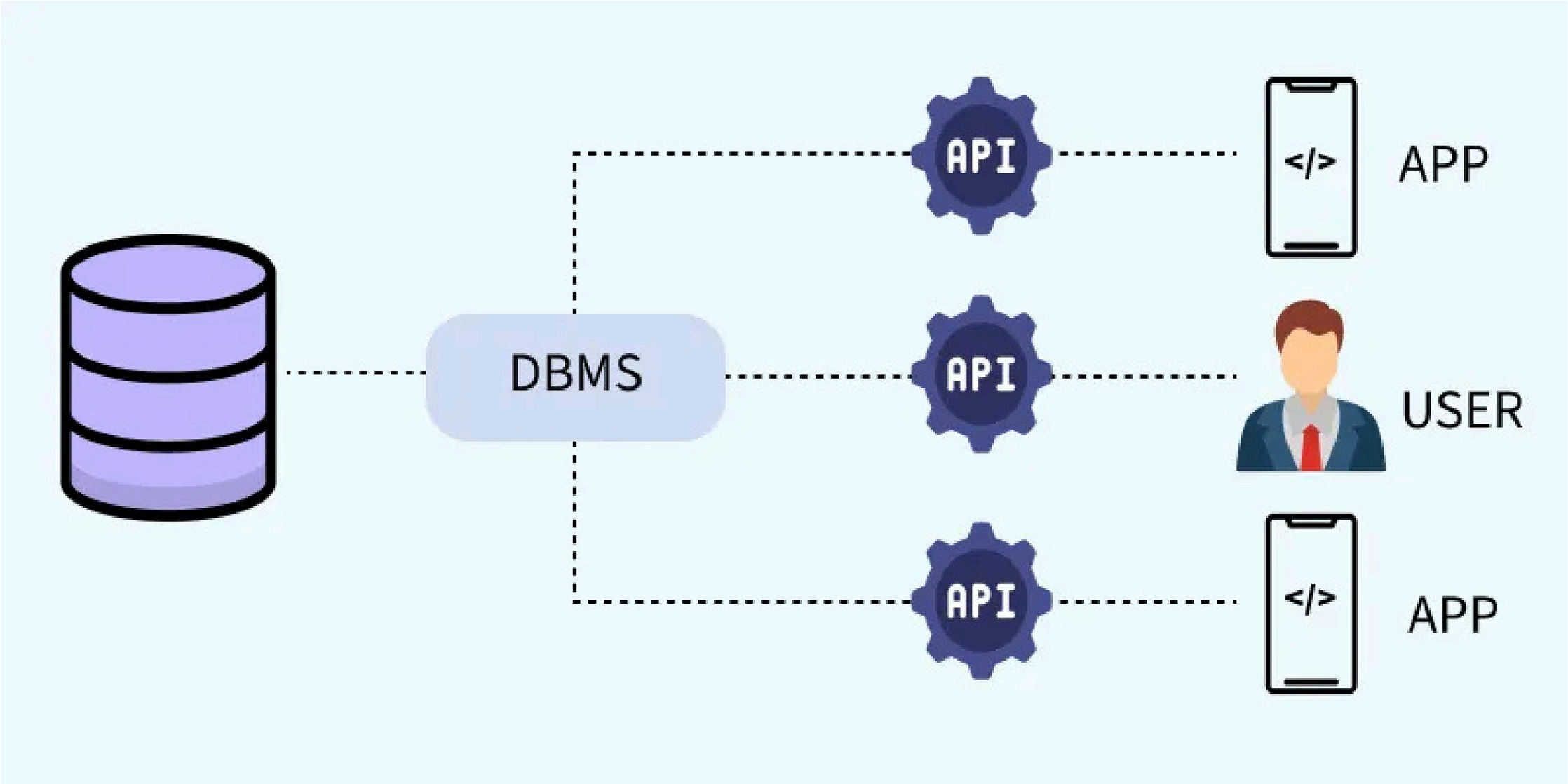
Learning objectives

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- Define what a database and a DBMS are, and explain their advantages over traditional file-based storage systems
- Classify different types of databases (relational, document, key-value, columnar, graph) and identify appropriate use cases for each
- Explain the three-schema architecture and the fundamental components of the relational model (tables, rows, columns, keys)
- Summarize the ACID properties (Atomicity, Consistency, Isolation, Durability) and explain their importance in reliable database systems
- Compare popular DBMS platforms (MySQL, PostgreSQL, Oracle, SQL Server, MongoDB) and distinguish the roles within the database ecosystem (DBA, Data Engineer, Analyst, Developer)
- Install and configure a DBMS and client tool, and create a simple database/table through hands-on practice

Introduction to DBMS

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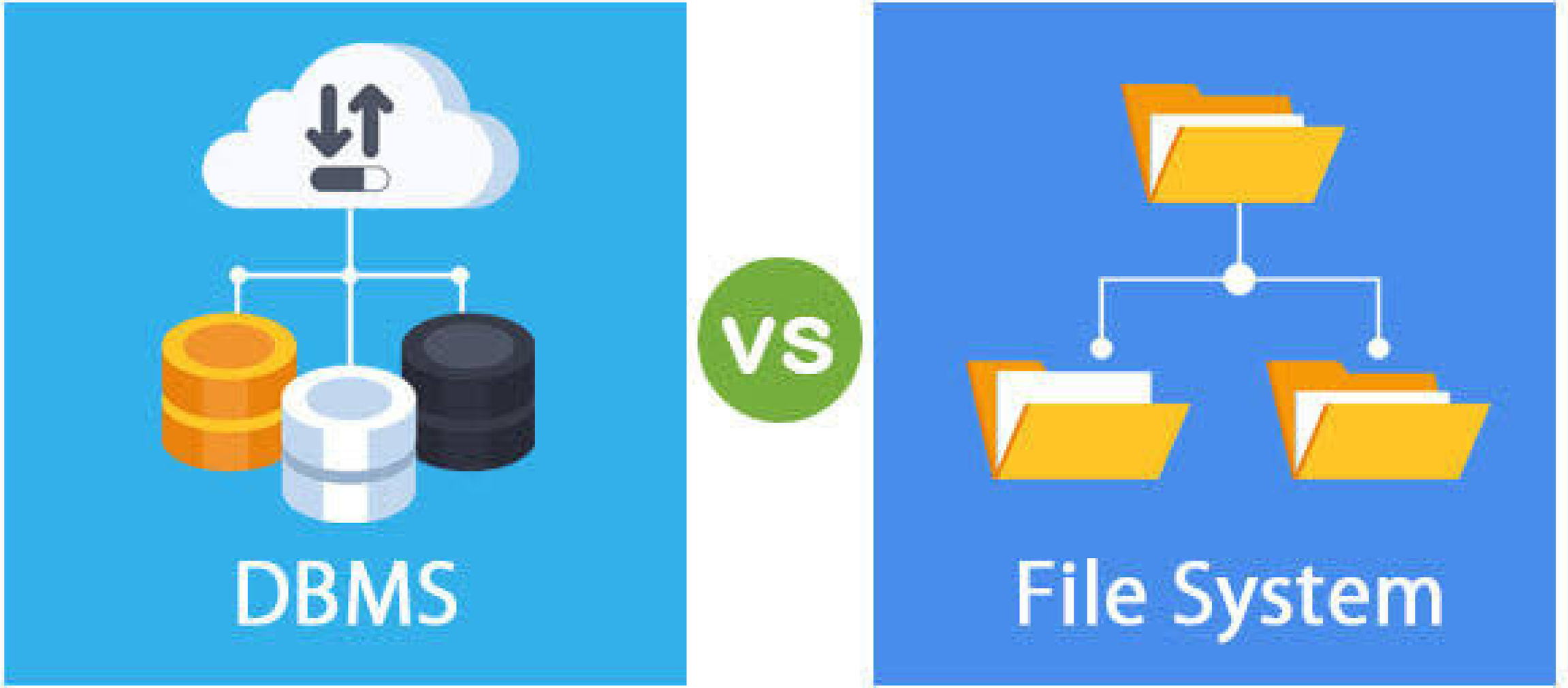
Introduction to DBMS

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- A database is an organized collection of structured data, stored and accessed electronically
- It allows data to be efficiently stored, managed, retrieved, and updated
- Databases are managed by a Database Management System (DBMS) — software that interacts with users, applications, and the database itself
- Examples of everyday databases:
  - Banking systems (accounts, transactions)
  - E-commerce platforms (products, orders, customers)
  - Social media platforms (users, posts, connections)
- Data vs. Information:
  - Data = raw, unprocessed facts (e.g., "25", "John")
  - Information = processed, meaningful data (e.g., "John, age 25")
- Why not just use files or spreadsheets?
  - Databases solve problems like data redundancy, inconsistency, and lack of security/concurrency control found in file-based systems

File-based systems vs. DBMS

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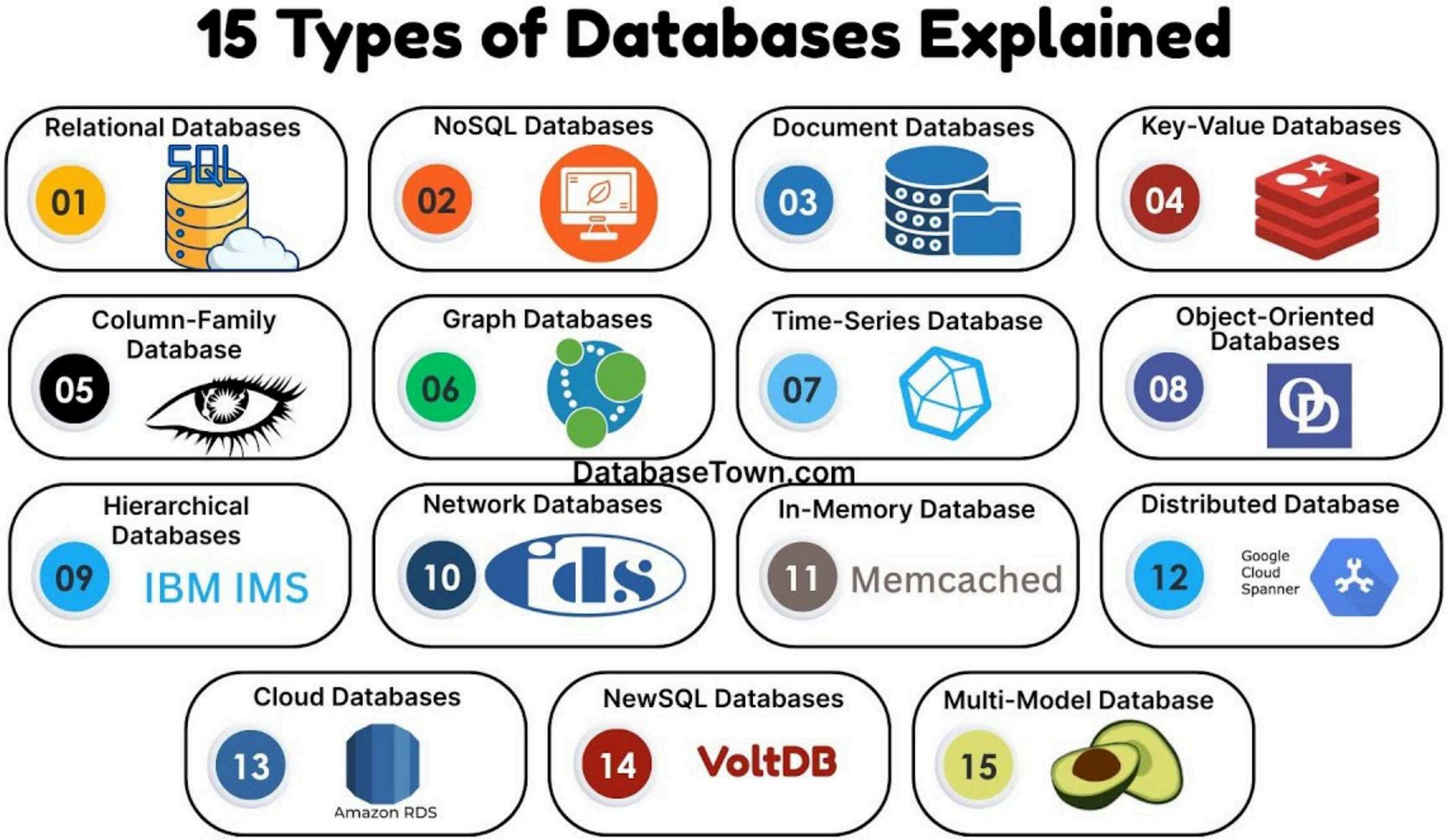
File-based systems vs. DBMS

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- File-Based Systems vs. DBMS
  - File-Based System: Data stored in separate files, managed directly by application programs (no centralized control)
- Problems with File-Based Systems:
  - Data Redundancy – Same data duplicated across multiple files
  - Data Inconsistency – Updates in one file not reflected in others
  - Difficulty in Accessing Data – Requires custom programs for every new query
  - Data Isolation – Data scattered across files in different formats
  - Integrity Problems – No built-in way to enforce rules (e.g., valid age ranges)
  - Concurrent Access Issues – Multiple users editing same file can cause conflicts/corruption
  - Security Issues – Hard to restrict access to specific data at a granular level
  - No Backup/Recovery Mechanism – Manual and unreliable

Types of DBs

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Types of DBs

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- Relational Databases (RDBMS)
  - Data stored in tables (rows & columns), related via keys
  - Uses SQL for querying
  - Examples: MySQL, PostgreSQL, Oracle, SQL Server
- NoSQL Databases – Designed for unstructured/semi-structured data and flexible scaling
  - Document Databases – Store data as JSON-like documents - MongoDB
  - Key-Value Databases – Store data as simple key-value pairs - Redis, DynamoDB
  - Columnar Databases – Store data by columns instead of rows, optimized for analytics - Cassandra, HBase
  - Graph Databases – Store data as nodes and relationships - Neo4j
- Hierarchical & Network Databases (legacy models)
  - Hierarchical: Tree-like parent-child structure (e.g., IBM IMS)
  - Network: Records connected via multiple links (e.g., IDMS)

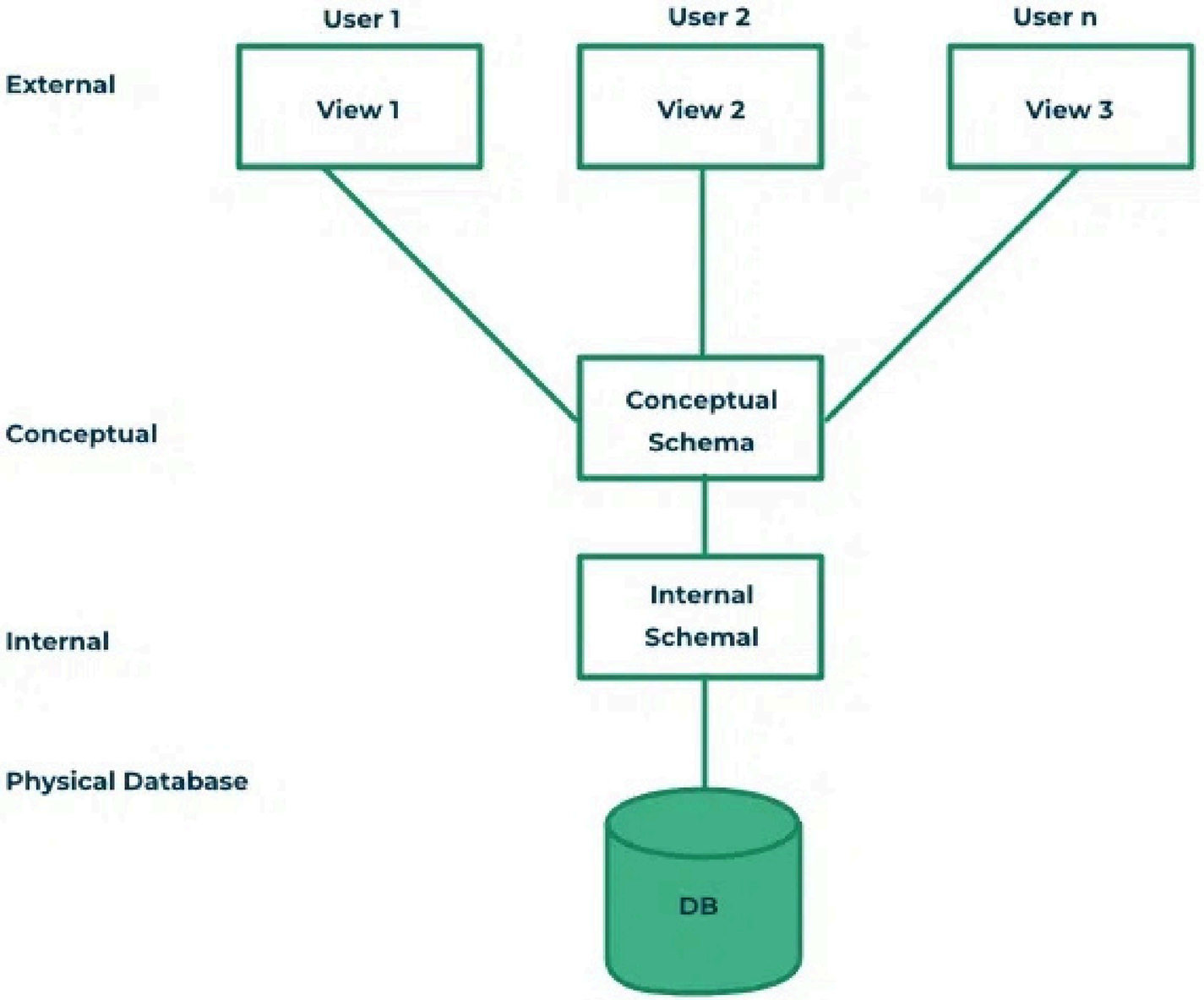
DBMS Architecture

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- Based on the Three-Schema Architecture, which separates how data is stored from how it's used
- External Level (View Level)
  - How individual users/applications see the data
  - Different users can have different views of the same database
- Conceptual Level (Logical Level)
  - Describes the overall structure of the database — entities, relationships, constraints
  - Independent of physical storage details
- Internal Level (Physical Level) - Describes how data is actually stored on disk (file structures, indexes, storage details)
- Data Independence
  - Logical Data Independence – Changing the conceptual schema without affecting external views
  - Physical Data Independence – Changing the internal schema without affecting the conceptual schema

DBMS Architecture

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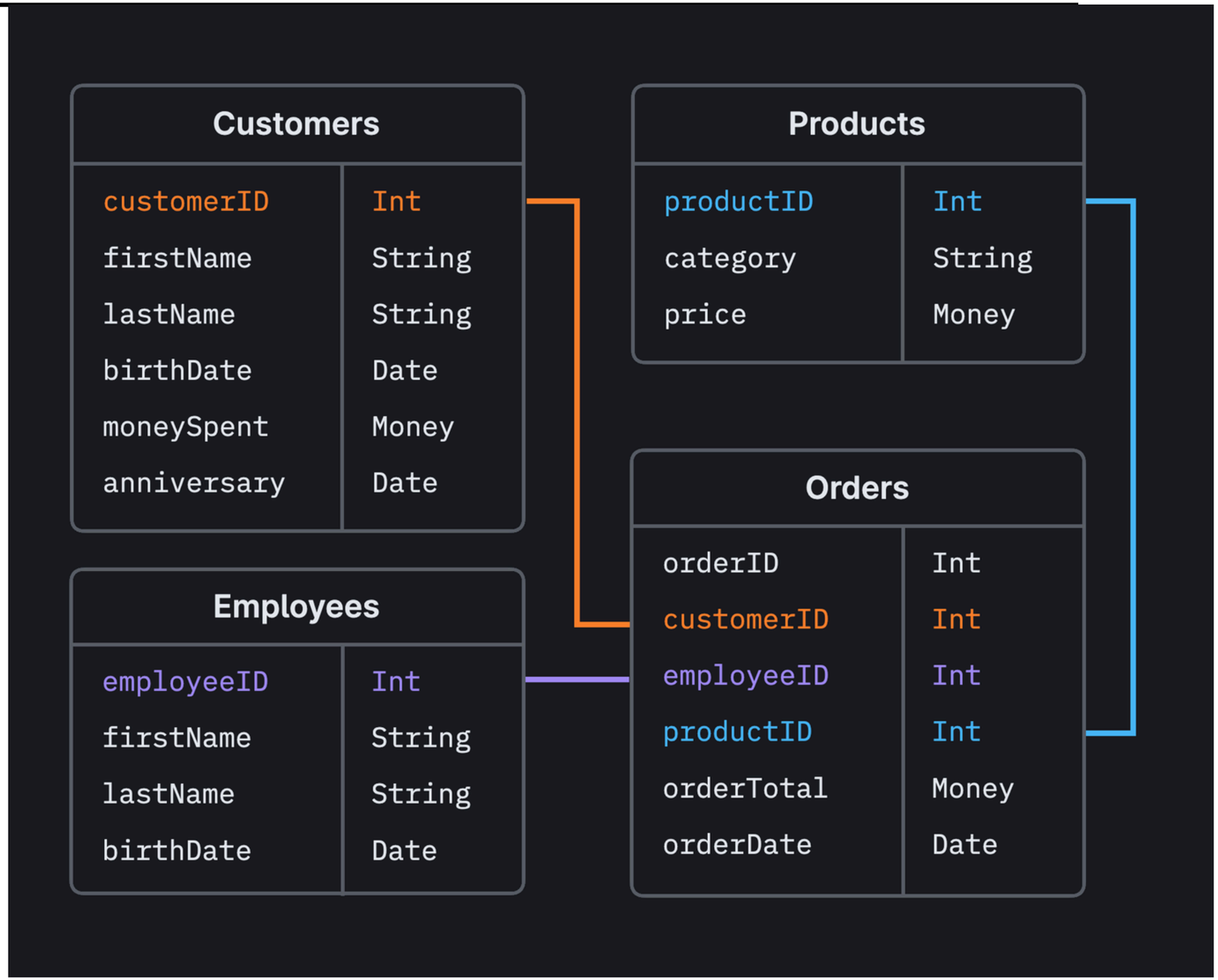
The Relational Model

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- Data is organized into tables (relations), consisting of rows and columns
  - Table (Relation): A collection of related data organized in rows and columns
  - Row (Tuple): A single record in a table, representing one entity instance
  - Column (Attribute): A single field/property of the entity (e.g., Name, Age, Email)
  - Domain: The set of allowable values for a given attribute (e.g., Age must be a positive integer)
  - Schema: The structure/blueprint of a table (column names and data types), independent of actual data
  - Instance: The actual data stored in a table at a given point in time
- Every table typically represents one entity type (e.g., Students, Courses, Employees)

The Relational Model

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Keys

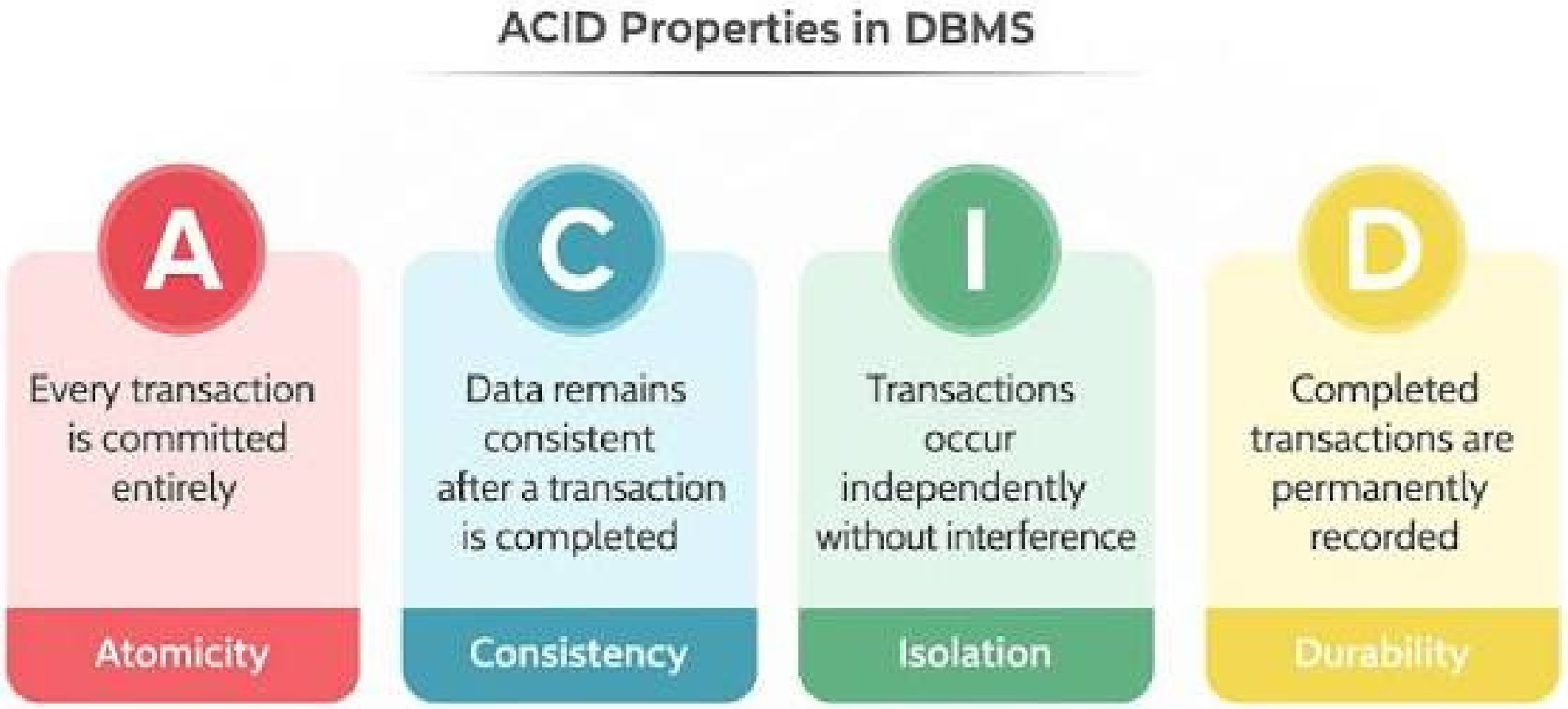
- Keys are used to uniquely identify rows and establish relationships between tables
  - Super Key: Any set of attributes that can uniquely identify a row
  - Candidate Key: A minimal super key — no unnecessary attributes
  - Primary Key: The chosen candidate key used to uniquely identify each row (cannot be NULL, must be unique)
  - Foreign Key: An attribute in one table that references the Primary Key of another table — used to establish relationships
  - Composite Key: A primary key made up of two or more attributes combined
- Keys enforce data integrity and enable relationships between tables

ACID Properties

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- A set of properties that guarantee reliable processing of database transactions
  - Atomicity
    - A transaction is all-or-nothing — either all operations complete, or none do
  - Consistency
    - A transaction brings the database from one valid state to another, maintaining all defined rules/constraints
  - Isolation
    - Concurrent transactions execute as if they were run one at a time, without interfering with each other
  - Durability
    - Once a transaction is committed, its changes are permanent — even in case of system failure
- ACID properties ensure data reliability, accuracy, and integrity in multi-user database systems

ACID Properties



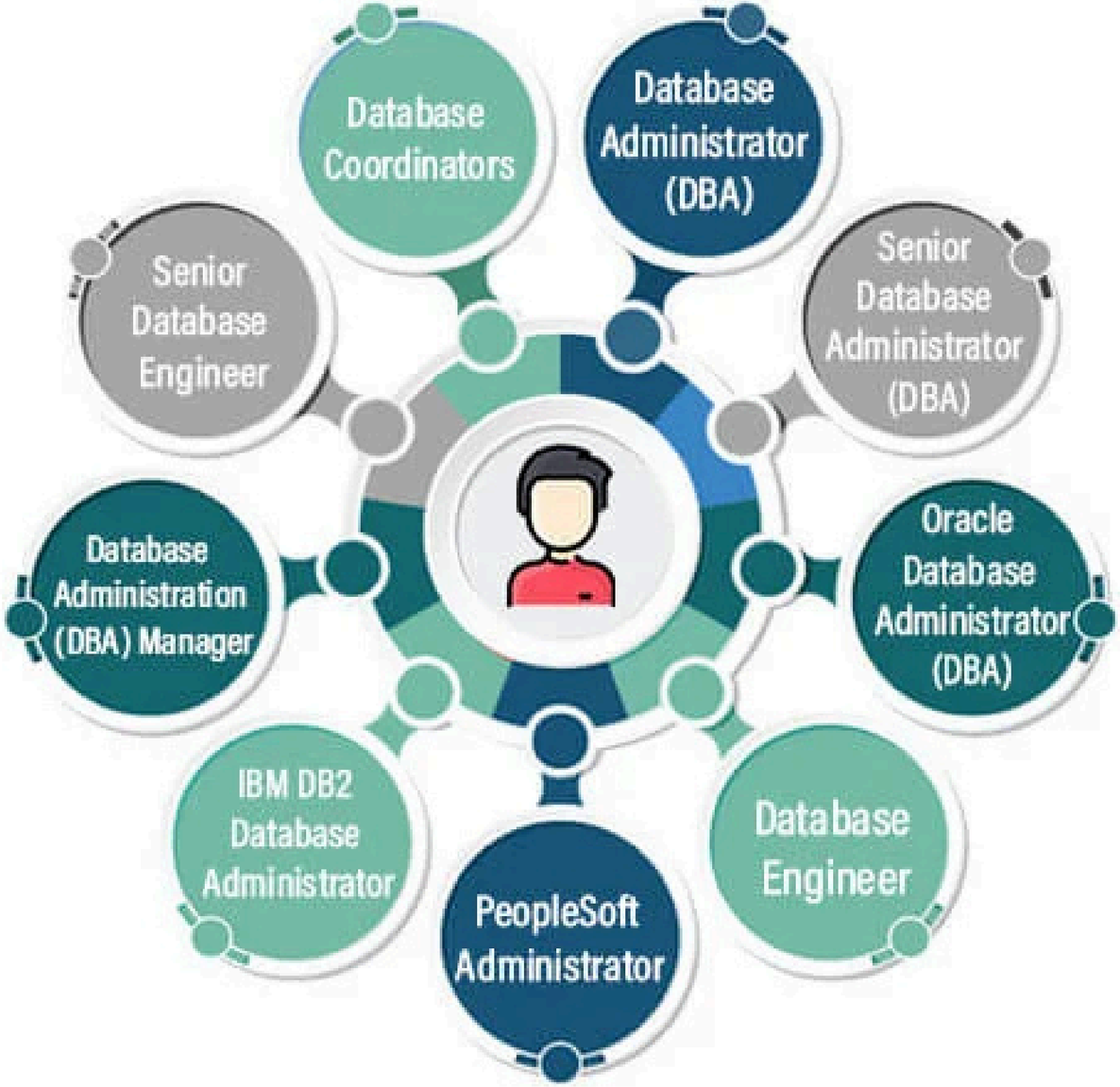
Popular DBMS Overview

- MySQL – Open-source, widely used for web applications, easy to learn
- PostgreSQL – Open-source, highly standards-compliant, supports advanced features (JSON, custom types)
- Oracle Database – Enterprise-grade, feature-rich, widely used in large organizations
- Microsoft SQL Server – Enterprise DBMS, tightly integrated with Microsoft ecosystem
- MongoDB – Popular NoSQL database, stores data as JSON-like documents
- Choosing a DBMS depends on:
  - Project size and complexity
  - Licensing costs (open-source vs. commercial)
  - Scalability and performance needs
  - Ecosystem/tool integration

Popular DBMS Overview



Roles in Database World



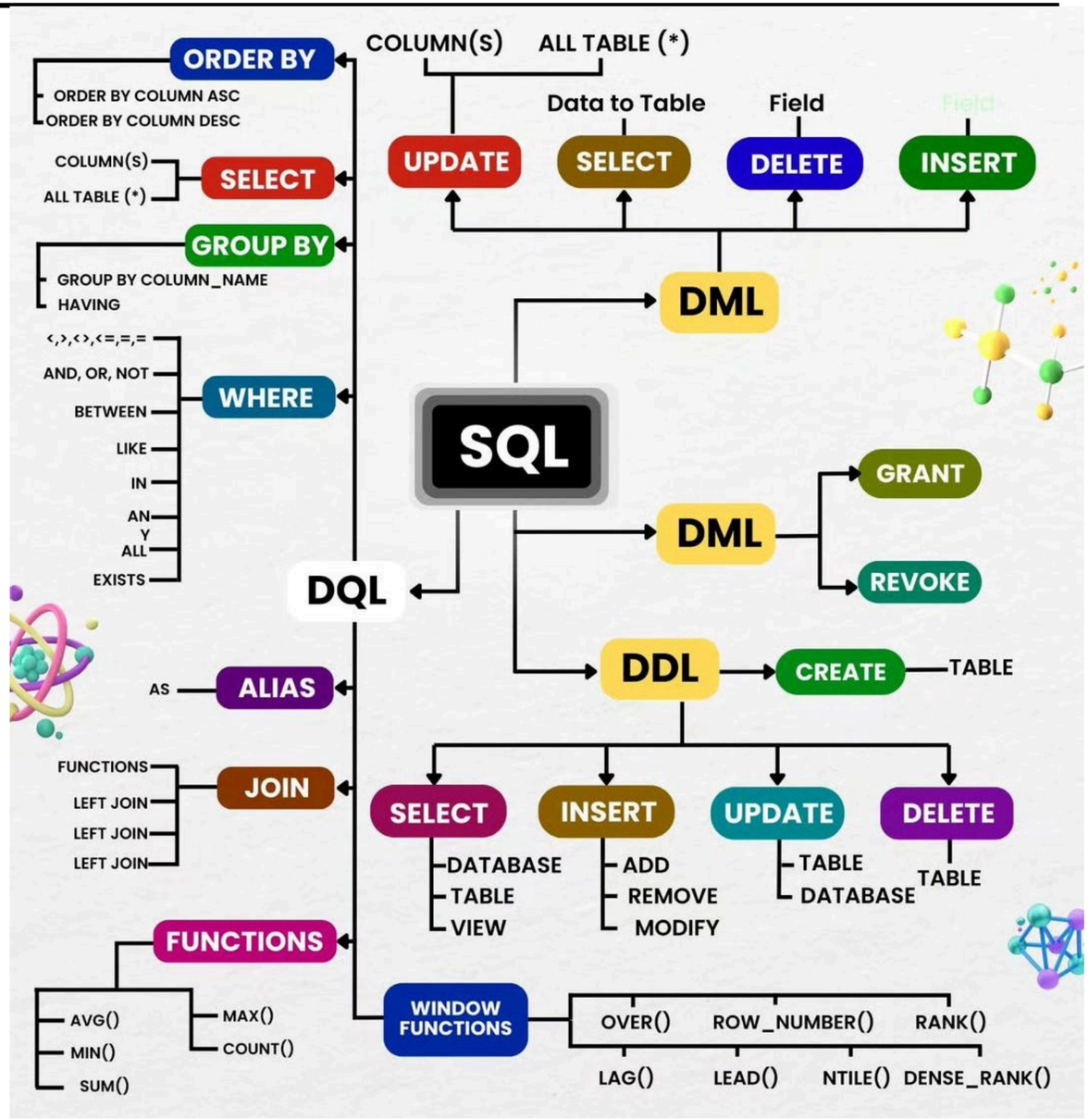
Roles in Database World

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- Database Administrator (DBA)
  - Manages database installation, security, backup/recovery, and performance tuning
- Data Engineer
  - Builds and maintains data pipelines, ETL processes, and data infrastructure
- Data Analyst
  - Analyzes data to generate insights, reports, and support decision-making
- Database Developer
  - Designs schemas, writes queries, stored procedures, and integrates databases with applications
- Key Difference:
  - DBA → maintains & secures the system
  - Developer → builds features using the system
  - Analyst → extracts insights from the system
  - Engineer → moves & transforms data at scale

Introduction to SQL

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Introduction to SQL

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- SQL (Structured Query Language) – Standard language used to communicate with relational databases
- Used to create, read, update, and delete data (CRUD operations)
- SQL Command Categories:
  - DDL (Data Definition Language) – Defines structure: CREATE, ALTER, DROP
  - DML (Data Manipulation Language) – Manages data: SELECT, INSERT, UPDATE, DELETE
  - DCL (Data Control Language) – Manages permissions: GRANT, REVOKE
  - TCL (Transaction Control Language) – Manages transactions: COMMIT, ROLLBACK
- SQL is declarative — you specify what you want, not how to get it
- Nearly all relational DBMS platforms use SQL (with slight syntax variations)

Lecture Summary

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- A database is an organized collection of data, managed through a DBMS
- DBMS solves major limitations of file-based systems: redundancy, inconsistency, poor security, no concurrency control
- Databases come in several types: Relational, Document, Key-Value, Columnar, Graph
- The Three-Schema Architecture separates external views, logical structure, and physical storage
- The Relational Model organizes data into tables, using keys to ensure uniqueness and relationships
- ACID properties guarantee reliable and consistent transactions
- Different roles (DBA, Developer, Analyst, Engineer) work with databases in different capacities
- Hands-on setup prepares students for writing SQL and managing databases starting next week

Hands-on Setup

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- Goal: Install a DBMS and client tool to prepare for hands-on labs
- Step 1: Install a DBMS: PostgreSQL
- Step 2: Install a Client Tool: pgAdmin
- Step 3: Verify Installation: Connect to the DBMS using the client tool
- Step 4: Create Your First Database: Run CREATE DATABASE command

Next Lecture

- SQL Fundamentals I (DDL, DML, basic queries)